# Practical: Flour, Formulas and the Protocol as Run

Part IV of the reference, the heat-driven levain bank, and a plan for maximum flavour

Sourdough Science

github.com/zrrbite/sourdough-science

# What this deck covers

1. **Flour** — why protein % is not enough; what to buy (ch. 21)
2. **Formulas and method** — ratios, two pizza formulas, seven steps (ch. 22)
3. **Quick references** — home oven vs Ooni 12”; five preferments (ch. 23–24)
4. **The protocol as run** — the heat-driven levain bank, its hypotheses and open tests
5. **A plan for maximum flavour** — seven steps aimed at acids, esters or the crust

Flags: **own result** this kitchen’s experience **hypothesis** proposed mechanism
**contested** the literature conflicts **not yet tested** open, not yet checked

**In the kitchen:** Formulas assume a domestic oven to 300°C, an Ooni 12” gas, Caputo Oro Manitoba, a pure sourdough culture kept warm, and a fridge at 3–5°C. Adjust for your own kit.

Source: README.md (equipment and conventions).

# Flour

# The protein number says how much, not how good

- Protein % on the bag is Kjeldahl nitrogen × a factor: the total, not the composition
- What matters: the glutenin-to-gliadin ratio and the **HMW glutenin subunit** alleles
- Glu-D1 **5+10**: longer, more cross-linked polymers — tighter mesh, more elasticity, better gas retention
- Glu-D1 **2+12**: shorter, weaker polymers — yet the same protein % on the bag
- European retail “Manitoba” is essentially unregulated, meaning roughly “high protein”: no origin or CWRS genetics guaranteed

**In the kitchen:** Two flours at the same protein % can make very different doughs. The label cannot tell you which.

Source: ch. 21; Payne 1987 (Glu-1 scoring, 5+10 vs 2+12); Shewry, Halford & Lafiandra 2003.

# Four numbers predict whether a dough survives 48 h

| Measure | What it tells you | Caputo Oro Manitoba | Generic flour |
|---|---|---|---|
| W index ($J \times 10^{-4}$) | Energy to inflate a dough sheet to bursting: strength + extensibility | W 350–380+ | “Manitoba” at 13%: often W 280–310 |
| P/L ratio | Tenacity vs extensibility; ideal ~0.5–0.7 | Formulated to hit 0.5–0.7 | High P: snaps back, tears. Low P: puddles on the peel |
| Damaged starch | Broken granules feed amylase: more sugar | Controlled by calibrated milling | Often more, from aggressive rollers |
| Ash | Minerals that activate amylases and proteases | ~0.45–0.55% (Tipo 00 spec) | 0.65–0.80%, even at the same protein |

**In the kitchen:** The 50–80 W gap decides whether a 48 h dough reaches the oven intact. More damaged starch or ash leaves more sugar after the retard: an Ooni burn risk.

Source: ch. 21; Chopin 1927 (alveograph, W index); Barrera et al. 2013 (damaged starch).

# For a long retard, buy flour by W, not protein

| Choice | Spec | Why / how to use it |
|---|---|---|
| First choice | Caputo Oro Manitoba / premium Tipo 00: W 320+, P/L 0.5–0.7, protein 13.5–14.5%, ash <0.55% | Made for long ferments; survives 48 h with the network intact |
| Acceptable | Any Tipo 00 or strong bread flour with verified W 300+ | No W given? Protein over 13.5% from a reputable mill: a fair, imperfect proxy |
| Use with caution | Generic “Manitoba” under 13% or unknown W: fine short, likely to degrade in 48 h | Retard 24 h, cut the levain (less acid, less protease), or blend in 20–30% Caputo Oro |
| Levain flour | Whole wheat or high-extraction | Minerals, flavour precursors, enzymes — kept to the levain’s 10–15% of the flour |

**In the kitchen:** Flour barely matters for a 2-hour direct dough; for a 48 h retarded hybrid in an Ooni, it is the highest-impact ingredient.

Source: ch. 21.

# Formulas and method

# Levain up, yeast down: three starting ratios

| | Conservative | Moderate | Bold |
|---|---|---|---|
| Levain (% of total flour) | 10% | 15% | 20% |
| IDY (% of total flour) | 0.3% | 0.2% | 0.15% |
| Levain hydration | 100% (liquid) | 100% (liquid) | 60% (stiff) |
| Levain maturity | Young, just peaked | Just peaked | Peaked, slight decline OK |
| Acid character | Subtle background tang | Noticeable lactic roundness | Distinct, complex tang |
| Gluten impact | Minimal | Slight softening | Moderate — may need stronger flour |
| Flavour in home oven | Mild enhancement | Clear sourdough character | Pronounced, rustic |

**In the kitchen:** More levain brings more wild yeast, so cut the IDY: at 20% with a mature culture, 0.1% may do. A stiff levain brings less water: add it to the dough.

Source: ch. 22 (home oven pizza, 250–300°C).

# The Ooni formula is drier and carries far less yeast

| | Home oven, 4 × ~250 g | | Ooni 12”, 4 × ~275 g | |
|---|---|---|---|---|
| **Ingredient** | **Weight** | **Baker’s %** | **Weight** | **Baker’s %** |
| Flour, W280–320 (Ooni: Tipo 00 only) | 550 g | 100% | 600 g | 100% |
| Water | 340 g | ~62% | 350 g | ~58% |
| Liquid levain (100% hydration, at peak) | 165 g | 15% flour equiv. | 150 g | ~12.5% flour equiv. |
| Fine sea salt | 16 g | 2.9% | 18 g | 2.7% |
| Instant dry yeast | 1.1 g | 0.2% | 0.4 g | 0.06% |
| Olive oil (optional) | 15 g | 2.7% | 12 g | 1.8% |
| True flour / water, incl. the levain’s | ~632 / ~422 g | **~67%** | ~675 / ~425 g | **~63%** |

**In the kitchen:** Ooni: 48 h minimum retard, non-negotiable: the yeast eats most of the sugar, so leopard spots, not burning. Under-proofed? 0.1% IDY next batch. Blown? Less, or retard longer.

Source: ch. 22. Home flour: Tipo 00 or strong bread flour.

# Steps 1–4: young levain, acid in late, folds early

| Step | When | Do | Why |
|---|---|---|---|
| 1. Build the levain | 8–12 h before mixing; room temperature | Feed 1:5:5 (culture:flour:water) with white bread flour, until just peaked: domed, bubbly, floats | Caught at or just before peak: most yeast vigour relative to LAB |
| 2. Autolyse | 30–60 min | Flour + most of the water (hold back ~30 g); mix until no dry flour; cover | Hydrates the flour and starts gluten before the levain's acid arrives |
| 3. Final mix | — | Add levain, the rest of the water with the yeast dissolved in it, salt, oil | Mix to smooth, slightly tacky, a moderate windowpane |
| 4. Bulk | 2–3 h at ~22° C | 3–4 stretch-and-fold sets, 30 min apart, in the first 90 min; then leave it | Expect +50–70% volume; the commercial yeast keeps timing predictable |

Source: ch. 22 (method for both formulas).

# Steps 5–7: cold retard, gentle pull, hot bake

| Step | When | Do | Why |
|---|---|---|---|
| 5. Ball, retard | 3–5°C. Home 24–72 h; Ooni 48–72 h minimum | Home 250 g; Ooni 270–280 g (thicker, forgiving) or 220–230 g (thinner, crisp). Round tightly; oiled tubs | LAB acidify slowly, yeast nearly stops, ester precursors build. Ooni: sugar depletion |
| 6. Temper, gentle pull | 1.5–2 h, room temperature | Stretch and fold each ball 2–3 times, turning 90°; re-round loosely, no degassing | $CO_2$ dissolved in the cold nucleates (Henry's law): visible rise in 20–30 min |
| 7a. Bake, home | 250–300°C, 6–10 min | Preheated stone or steel | Longer bake, more roasted, malty crust aroma (ch. 14) |
| 7b. Bake, Ooni | 350–380°C stone | Preheat full flame ~15 min, then 40–50%. 270–280 g: 10–11", 2–3 min, turn every 20–30 s. 220–230 g: 11–12", 90 s–2 min | Optional final 15–20 s boost for leopard spots; 3–5 min stone recovery at full flame between pizzas |

Source: ch. 22 (method); ch. 19 for the gentle pull.

# Quick references

# Sugar is fuel in the home oven, a burn risk in the Ooni

| Parameter | Home oven (250–300°C) | Ooni 12” (350–380°C stone) |
|---|---|---|
| Total hydration | 65–68% | 63–65% |
| Ball weight / pie diameter | ~250 g / 12” | ~275 g / 10–11” |
| Commercial yeast (IDY) | 0.2% | 0.05–0.1% |
| Levain % | 15% (moderate) | 10–15% |
| Cold retard | 24–72 h (flavour) | 48–72 h (mandatory for sugar depletion) |
| Flame setting | N/A (thermostat) | 40–50% of max (preheat at full, drop before launch) |
| Bake time / stone recovery | 6–10 min / N/A | 2–3 min + optional 15–20 s char boost / 3–5 min at full flame |
| Primary burn risk | Low — more browning fuel is welcome | High — residual sugar carbonises at max flame |
| Raw centre risk | Very low (long bake) | High at max flame — moderate flame solves this |
| Crust aroma (Maillard) | Extended thermal phase, rich expression | Good at 2–3 min; compressed at 60 s |
| Key advantage of hybrid | Flavour complexity in longer bake | Sugar depletion + flavour + cooked-through centre |

Source: ch. 23.

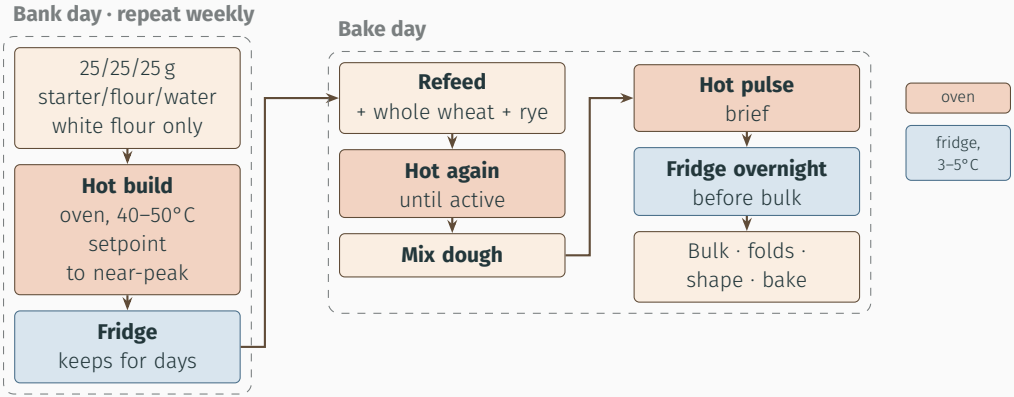
# Hydration and organism set acid, flavour and gluten

| | Biga | Poolish | Stiff levain | Liquid levain | Hybrid (yeast + levain) |
|---|---|---|---|---|---|
| Hydration | 50–60% | 100% | 50–60% | 100% | Any (levain separate) |
| Microbiology | *S. cerevisiae* | *S. cerevisiae* | Wild yeast + LAB | Wild yeast + LAB | *S. cerevisiae* + LAB |
| Primary acid | Minimal | Minimal | Acetic (sharp) | Lactic (mild) | Tuneable (levain type) |
| pH range | 5.5–6.0 | 5.5–6.0 | 3.5–4.5 | 3.8–4.5 | 4.5–5.5 |
| Flavour | Yeast aromatics | Sweet, wheaty | Sharp, vinegary | Round, creamy | Complex esters, balanced tang |
| Gluten effect | Elastic, tight | Extensible, soft | Extensible + protease | Extensible + protease | Mostly elastic, slight acid softening |
| Crumb | Fine, uniform | Open, regular | Wild, irregular | Open, moderate | Controlled open, uniform |
| Ferment time | 12–24 h | 8–16 h | 6–12 h (cycle) | 4–8 h (cycle) | 2–3 h bulk + cold retard |
| Best for | Ciabatta, focaccia | Baguettes, Neapolitan pizza | Country bread | Enriched doughs | Home oven pizza |
| Pizza suitability | Good | Excellent (traditional) | Viable | Good | Optimal for home oven |

Source: ch. 24 — *"Same vessel, same hydration, completely different ecology. Or: combine them."*

# The protocol as run

# Hot, cold for days, hot, hot, cold overnight, then bulk




**In the kitchen:** Bank several jars at once; on bake day, one jar refed with wholegrain is the levain. This is how the bread is made here now **own result**; why it works is still hypothesis.

Source: notes/heat-driven-levain-bank.md.

# What changed: the cold rest now comes before bulk

| | Chapter 08 protocol | This protocol |
|---|---|---|
| Levain | Built hot, used at once | Built hot, banked cold for days, refed hot |
| Wholegrain | In the build | Withheld until the bake-day refeed |
| Hot dough phase | 2 h bulk at 40°C+ | Brief pulse straight after mixing |
| Cold phase | After bulk (retard) | Before bulk (enzymatic rest), optionally also after |
| Bulk | Hot | Room temperature, after the cold rest |

- The cold-before-bulk step is the one that changes the character of the result **own result**
- The idea, from the *Double Enzymatic Activation* notes: amylase works overnight while the microbes are dormant, so the dough reaches bulk pre-sweetened and extensible
- Here that rest is stacked on a hot front end instead of replacing it

Source: notes/heat-driven-levain-bank.md.

# Every hot session follows a fresh feed and a rest

**hypothesis** a mechanism proposed in the note, not yet tested

- Ch. 11: the most useful defence against heat stress is to **feed right before a hot session** — most substrate, least acid and ethanol, a full ATP budget
- This protocol does it at every hot step without being designed to: the bank build, the bake-day refeed, and the dough mix (the largest feed of all)
- Ch. 11 frames safe heat as brief excursions with recovery between; days at 3–5° C between build and refeed are a long recovery window
- So each hot pulse meets a culture that is rested, not depleted

**In the kitchen:** If you heat a levain, feed it just before. A ripe, sour one carries the acid and ethanol that make heat more lethal.

Source: notes/heat-driven-levain-bank.md; ch. 11.

# White flour in storage, wholegrain at activation

**hypothesis** a mechanism proposed in the note, not yet tested

## In the fridge: keep it inert

- The enemy is over-acidification
- Acid and ethanol make the *next* heat exposure more lethal (ch. 11)
- White flour has less amylase, fewer minerals and less LAB fuel, so the jar sours more slowly

## On bake day: load it

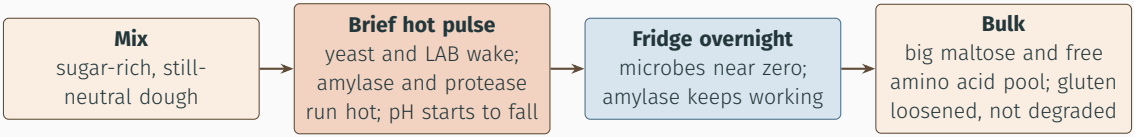
- The goal flips: most enzyme fuel, most microbial kick
- Whole wheat and rye: amylase, protease, minerals, lysine-rich protein for the crust
- Fuel arrives just before two hot pulses and the overnight rest, not burnt off in the fridge

**In the kitchen:** Test: bank one jar with 10% rye and one white-only; refeed and heat side by side; compare smell, rise and heat tolerance. **not yet tested**

Source: notes/heat-driven-levain-bank.md.

# Hot pulse for the enzymes, fridge to still the microbes

**hypothesis** a mechanism proposed in the note, not yet tested




- Enzyme rate roughly doubles per 10°C ($Q_{10}$): a dough at pH 5.5 and 40°C has proteolytic throughput comparable to one at pH 4.5 and 25°C
- $Q_{10} \approx 2$ is a general rule of biochemistry, an order-of-magnitude guide, not a dough measurement
- A plausible reason the dough comes out feeling “truly enzymatically activated” **own result**

Source: notes/heat-driven-levain-bank.md; notes/proteolysis-ph-heat.md; ch. 25 (note on $Q_{10}$).

# The risk is a slack dough, and four levers limit it

- An early cold rest relies on a high pH: protease stays quiet, only amylase works
- A hot pulse before the fridge erodes that: heat partly substitutes for low pH in driving protease
- The failure mode: the dough leaves the fridge slack rather than extensible

| Brief pulse | Low levain | Strong flour | Salt at the mix |
|---|---|---|---|
| Ch. 07: no more than 30–60 min for the dough itself | 10–15% | W 350+ | Salt is the largest single inhibitor of cereal protease |

**In the kitchen:** Here the overnight rest beats going straight to bulk **own result**, which suggests the balance is on the right side — with this flour.

Source: notes/heat-driven-levain-bank.md; ch. 07; notes/proteolysis-ph-heat.md.

# The oven air is 45°C; the levain core is still unknown

- “Lots of heat” is an oven setpoint, not the dough temperature (ch. 10)
- 75 g of levain in a glass jar ramps slowly in air; its core likely sits well below the setpoint for most of the session **hypothesis**
- A slow ramp protects better than a sudden shock (ch. 11): that could reconcile the protocol with ch. 07’s 40–45°C viability-decline range
- **Measured 2026-09-13:** empty-oven air ~45°C on the usual setting **own result**. A spot reading, not a logged hold, and not a loaded oven
- 45°C is where ch. 07 moves from “viability decline” to “thermal kill territory for most yeast”, so the explanation must come from the core

**In the kitchen:** The honest statement: the air is measured at 45°C, the core temperature is unknown, and the bread rises.

Source: notes/heat-driven-levain-bank.md; ch. 07, 10, 11; Guyot et al. (slow ramps); Sadot et al. 2017 (core lag).

# A plan for maximum flavour

# Flavour has three sources; each step works on one

| **Acids** from the bacteria | **Esters** from the yeast | **Crust browning** in the oven |
|---|---|---|
| **1** Whole-grain levain, stiffer for sharpness | **4** Keep the yeast active; handle gently late on | **3** Warm autolyse, no salt |
| **2** More levain in bread: 20–30% | **+** A pinch of commercial yeast | **5** Cold retard the shaped loaf |
| | | **6** Bake darker than feels right |
| | | **7** Eat the crust the same day |

**In the kitchen:** Change one thing per bake and taste it against your usual loaf. Several of these steps are bets, not measurements.

Source: the flavour plan, synthesised from ch. 04, 07, 12–14, 16, 17.

# Steps 1–2: a stiffer, bigger levain for more acid

## 1. Levain: whole grain, and stiffer if you want sharpness

- Rye or wholemeal in the bake-day feed, as the levain bank already does
- A stiff levain (50–60% water) shifts the acid toward acetic, which is sharper (ch. 04)
- Not to extremes: enough acetic acid stops sourdough yeast growing altogether — complete inhibition at 150 mmol/L undissociated acetic acid — and the yeast makes the esters

## 2. More levain in bread: 20–30%

- More acid, and more of the bacteria's protein-cutting enzymes; commits to a sourer loaf (ch. 17)
- Tension: the levain bank holds levain to 10–15% so a hot-pulsed dough won't go slack

**In the kitchen:** Raise levain or stiffness one bake at a time. Pizza ratios stop at 20% (ch. 22); 20–30% is for bread.

Source: ch. 04, 17; Gänzle et al. 1998 (acetic acid and sourdough yeast); notes/heat-driven-levain-bank.md.

# Steps 3–4: free amino acids early, keep the esters in

## 3. Warm autolyse, no salt: 60–90 min at 38–42°C, then add levain and salt

- Frees amino acids, which become the crust's malty, roasted aroma (ch. 17) **not yet tested**
- The levain bank's hot pulse and overnight cold rest do the same job **hypothesis**

## 4. Keep the yeast active, and handle gently late on

- Esters are made while the yeast works (ch. 13)
- A warm bulk may raise them, but the evidence conflicts: check by taste (ch. 07) **contested**
- Fold early, handle gently late: escaping $CO_2$ probably carries esters away (ch. 16) **hypothesis**
- A pinch of commercial yeast adds a large ester-making population without losing the rise (ch. 12–14)

Source: ch. 07, 12–14, 16, 17; Calvel et al. 2001 (autolyse); Kucharczyk & Tuszyński 2018 vs Ferrández-Gómez et al. 2024 (esters and heat).

# Steps 5–7: retard cold, bake dark, eat it the same day

## 5. Cold retard the shaped loaf: 12–36 h at 3–5°C

- Flour enzymes keep making sugar and amino acids for the crust, and the acid keeps building (ch. 14)
- Stop before the sugar runs out: sugar is what browns

## 6. Bake darker than feels right

- Most of a loaf's aroma forms in the bake (pumpernickel, Majcher et al. 2020); acid slows browning
- Steam for about 20 min at 250°C, then bake uncovered until deep brown **not yet tested**

## 7. Eat the crust the same day

- The roasty crust aroma is believed to fade within hours **not yet tested**; the sourness holds

Source: ch. 13, 14; Majcher et al. 2020. The bake times and temperatures are general practice, not from the repo.

# Takeaways

- Buy flour by W and P/L, not protein %: for a 48 h retard, W 320+ and P/L 0.5–0.7
- Levain up, yeast down: pizza takes 10–20% levain, bread tolerates 20–30%
- In the Ooni, a 48 h minimum retard uses up the sugar that would burn; a 40–50% flame cooks the centre
- Hydration and organism pick the preferment: stiff is acetic, liquid is lactic, the hybrid is tuneable
- The levain bank — hot builds after every feed, a cold rest before bulk — works here **own result**; why it works is still hypothesis, and the core temperature is unmeasured
- Flavour is acids, esters and crust: change one thing per bake and taste

# Open questions

## The levain bank not yet tested

1. Log core temperature in a banked jar and in the dough during the hot pulse
2. Bank life: how many fridge days before the refeed lags?
3. Does the hot build double? Doubling shows gas, not cell division; serial hot passage decides
4. 10% rye vs white-only at storage
5. With and without the overnight rest
6. Hot pulse of 30, 60, 90 min: where does extensible turn slack?

## Flavour and wider threads

- Warm bulk and esters: taste side by side against a room-temperature control (ch. 07) contested
- The Strecker/pyrazine crust claim: the least directly evidenced in the repo; wants a sensory comparison (ch. 17)
- The plan's bake times, and how fast crust aroma fades not yet tested
- Every temperature here is still a setpoint, not a reading: the logging chamber (`chamber/`) is designed, first stage built, not yet instrumented

Source: notes/heat-driven-levain-bank.md (open tests); README.md (status and open threads); the flavour plan.

# Sources


**Flour**

Payne PI (1987). Genetics of wheat storage proteins and the effect of allelic variation on bread-making quality. *Annu Rev Plant Physiol* 38:141–153.

Shewry PR, Halford NG, Lafiandra D (2003). Genetics of wheat gluten proteins. *Adv Genet* 49:111–184.

Chopin M (1927). Determination of baking value of wheat by measure of specific energy of deformation of dough. *Cereal Chem* 4:1–13.

Barrera GN et al. (2013). Effect of damaged starch on the rheological properties of wheat starch suspensions. *J Food Eng* 116(1):233–239.

**Heat and the levain**

Guyot S, Ferret E, Gervais P. Thermal death rate of *Saccharomyces cerevisiae*: dependence on the kinetics of heat perturbation.

Sadot M et al. (2017). Numerical modelling of convective heat transfer during bread baking. *J Food Eng*.

**Flavour**

Calvel R, Wirtz RL, MacGuire JJ (2001). *The Taste of Bread*. Aspen.

Majcher MA et al. (2020). Formation of key aroma compounds during preparation of pumpernickel bread. *J Agric Food Chem* 68(38):10352–10360.

Kucharczyk K, Tuszyński T (2018). The effect of temperature on fermentation and beer volatiles at an industrial scale. *J Inst Brew* 124(3):230–235.

Ferrández-Gómez B et al. (2024). Sourdough yeast strains exhibit thermal tolerance, high fermentative performance, and a distinctive aromatic profile in beer wort. *Foods* 13.

Gänzle et al. (1998). Cited in the flavour plan for 150 mmol/L acetic acid; not yet in ch. 25.


**In the repo**

docs/21–25; notes/heat-driven-levain-bank.md; notes/proteolysis-ph-heat.md; the flavour plan (from ch. 04, 07, 12–14, 16, 17).